



ECEN 5793

# Digital Image Processing

**Nate Lannan**

# Agenda

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Introduce sources of images

Introduce the human visual system

Discuss human perception

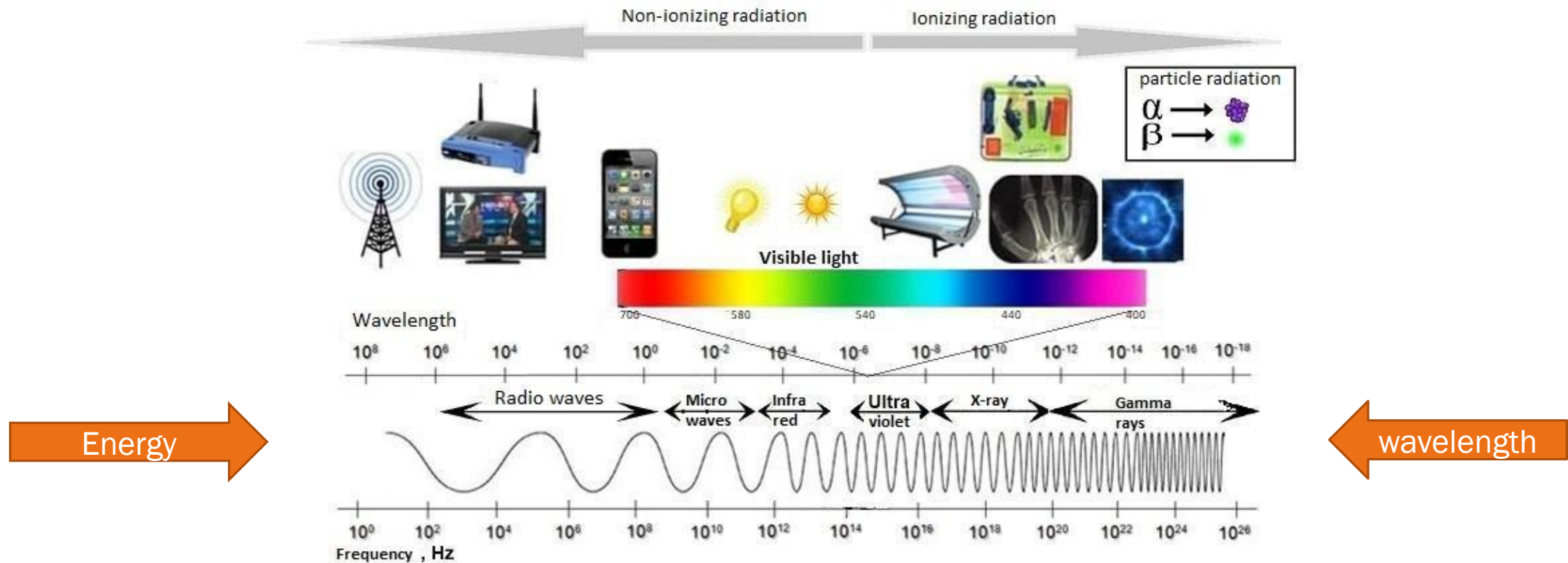
# How are images captured from different sources?

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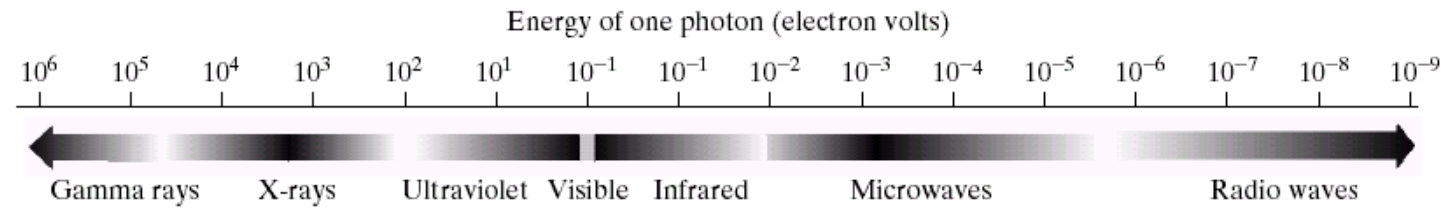


# Imaging Sources: Electromagnetic Radiation

## The electromagnetic spectrum



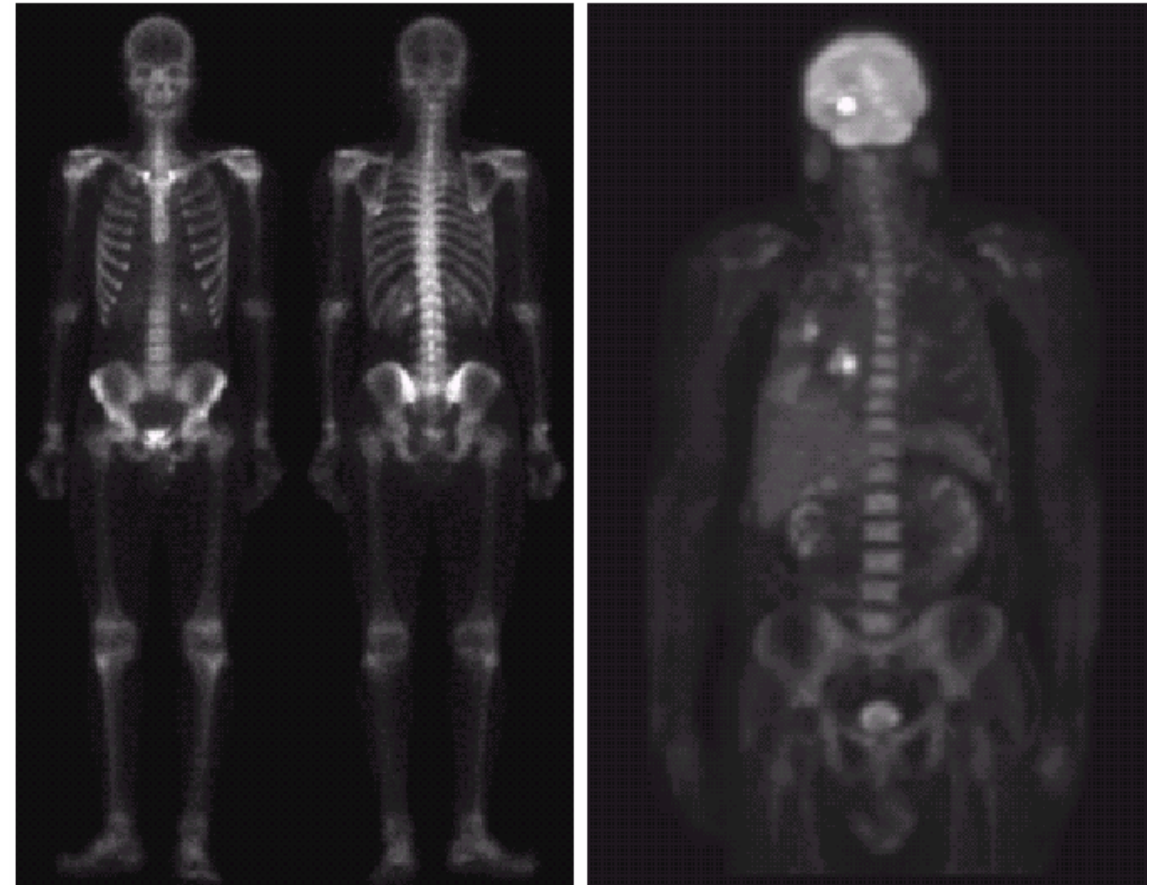
# Gamma-Ray Imaging



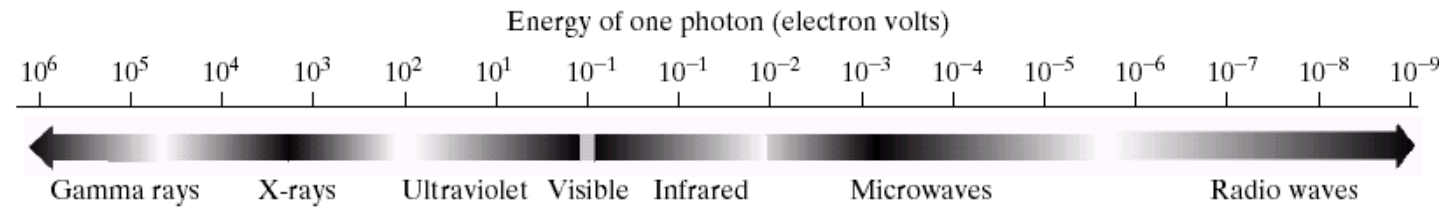
**FIGURE 1.5** The electromagnetic spectrum arranged according to energy per photon.

Major uses of imaging based on Gamma ray include nuclear medicine and astronomical observations.

- Inject a patient with a radioactive isotope that emits gamma-ray as it decays.
- Images are produced from the emissions collected by gamma ray detectors
- Gamma-ray imaging is widely used to diagnose various conditions and examine the structure and function of many different parts of the body, including the heart, brain, lungs, and kidneys



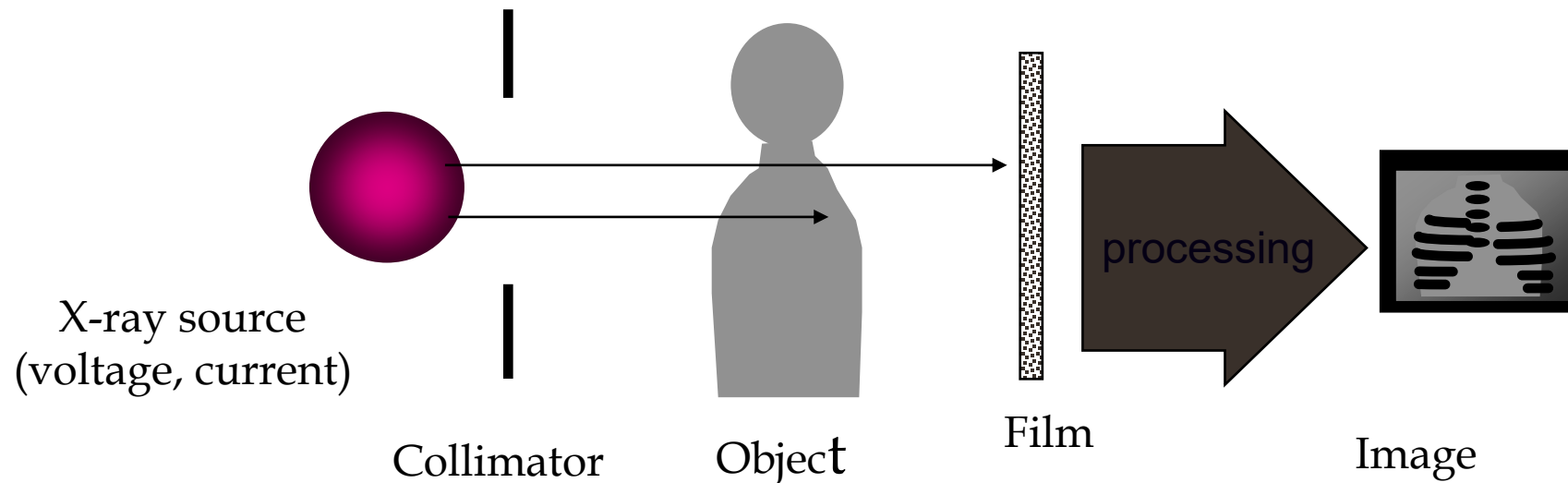
# X-Ray Imaging

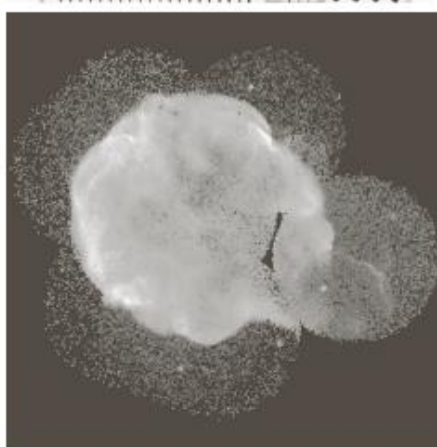
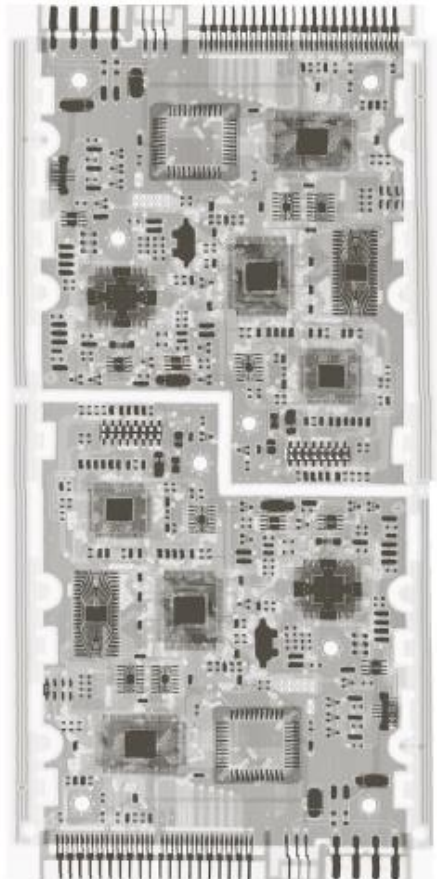


**FIGURE 1.5** The electromagnetic spectrum arranged according to energy per photon.

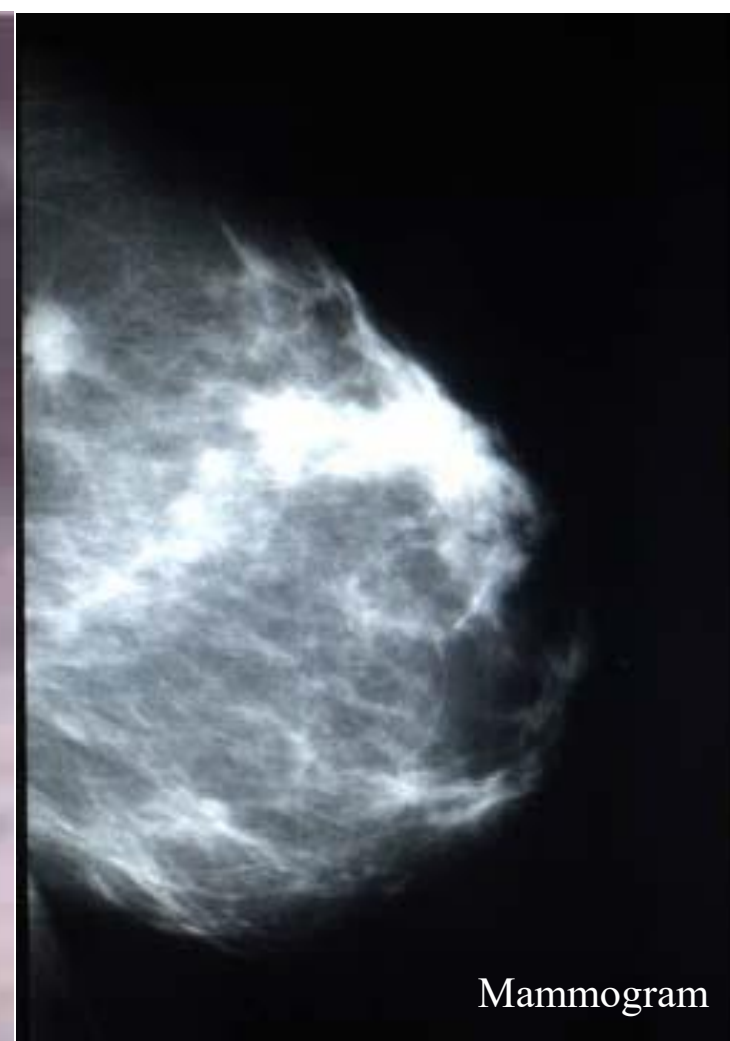
X-ray are among the oldest source of EM radiation used for imaging.

- Medical diagnosis
- Industry inspection
- Airport baggage inspection





Dental X-ray

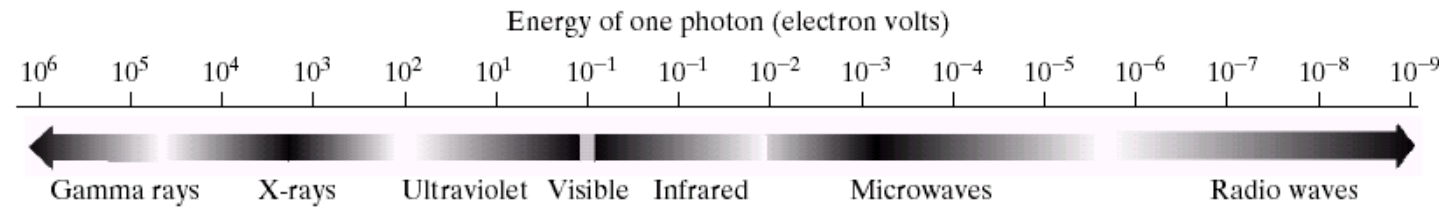


Mammogram

|   |   |
|---|---|
| a | d |
| b |   |
| c | e |

**FIGURE 1.7** Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center; (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School; (d) Mr. Joseph E. Pascente, Lixi, Inc.; and (e) NASA.)

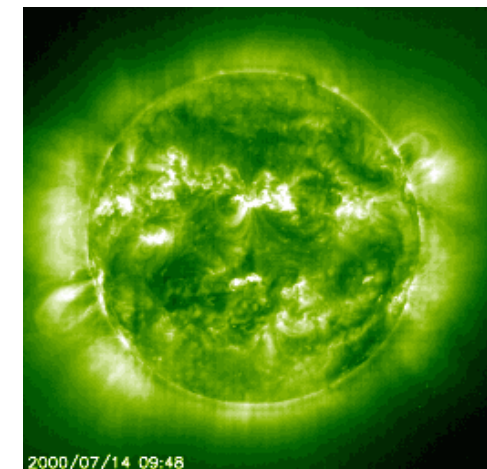
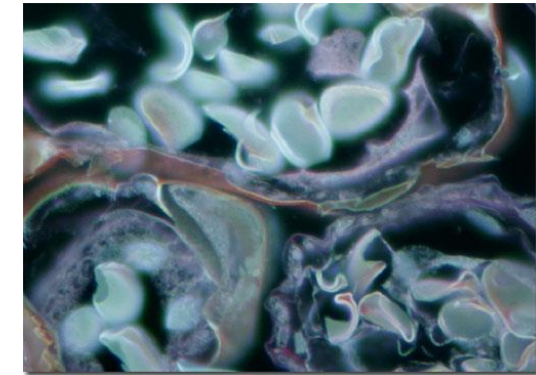
# Ultraviolet Imaging



**FIGURE 1.5** The electromagnetic spectrum arranged according to energy per photon.

Applications of ultraviolet “light” can be found in microscopy, lasers, biological imaging and astronomical observation.

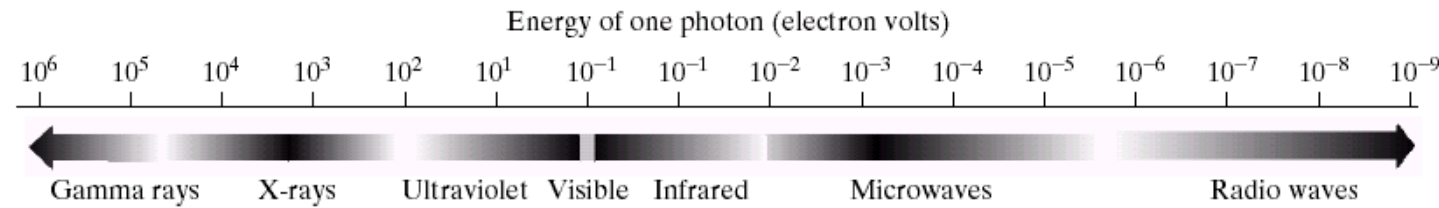
- Ultraviolet light can be used in microscopy to study fluorescent materials that absorb invisible light but emit visible light with a longer wavelength and a lower energy.
- The extreme ultraviolet imaging telescope (EIT) is an instrument on the space craft used to obtain high-resolution images of the solar corona.



[http://en.wikipedia.org/wiki/Image:Flare\\_EIT195\\_256px.gif](http://en.wikipedia.org/wiki/Image:Flare_EIT195_256px.gif)

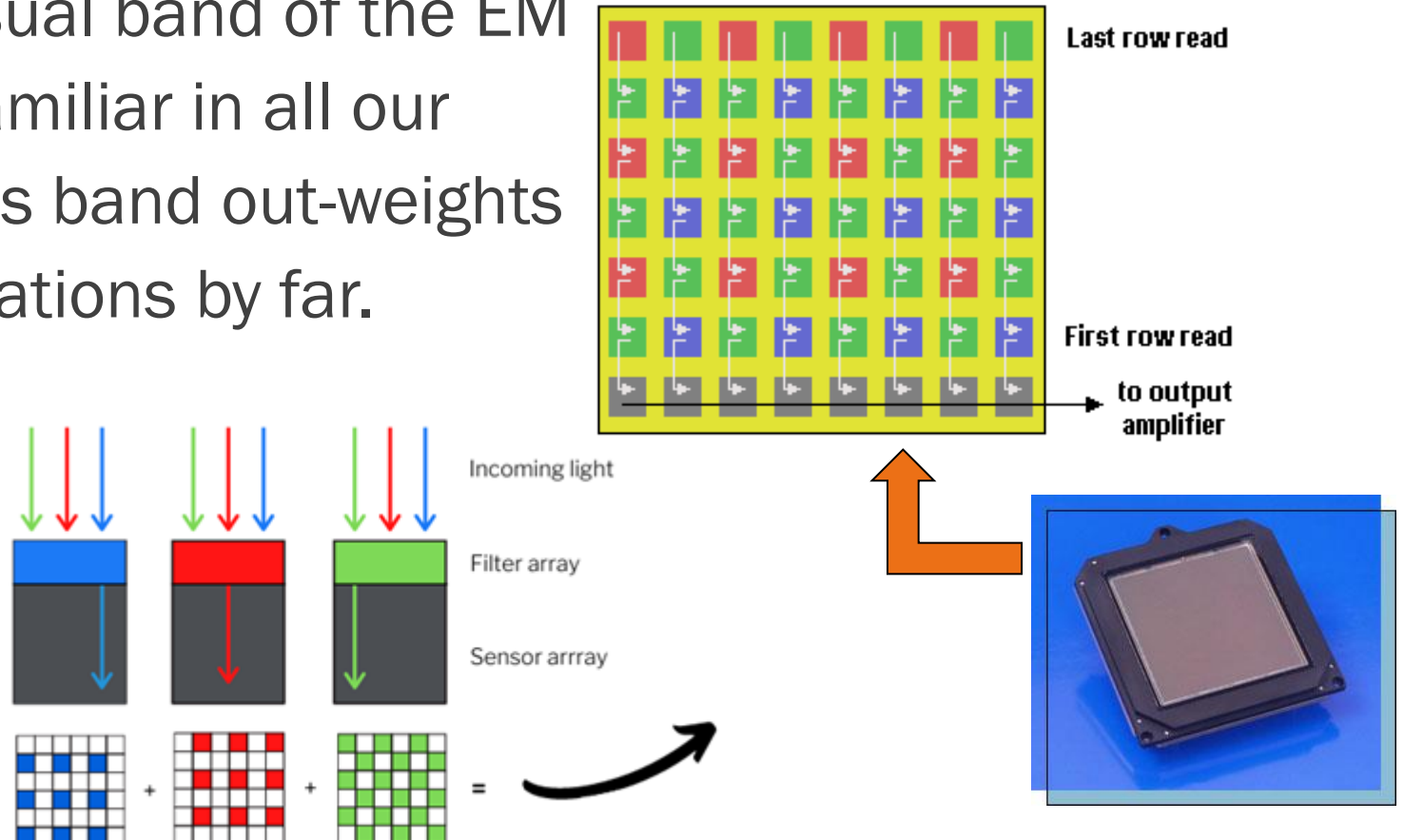
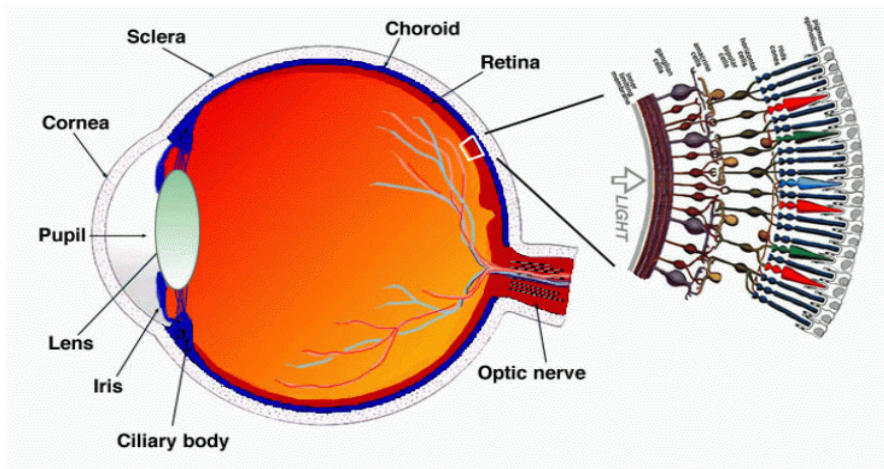
<http://www.microscopyu.com/articles/fluorescence/filtercubes/ultraviolet/uv2a/uv2acrytomiumlarge.html>

# Visible Band Imaging



**FIGURE 1.5** The electromagnetic spectrum arranged according to energy per photon.

Considering that the visual band of the EM spectrum is the most familiar in all our activities, imaging in this band out-weights all other imaging applications by far.



# Infrared Imaging Remote Sensing

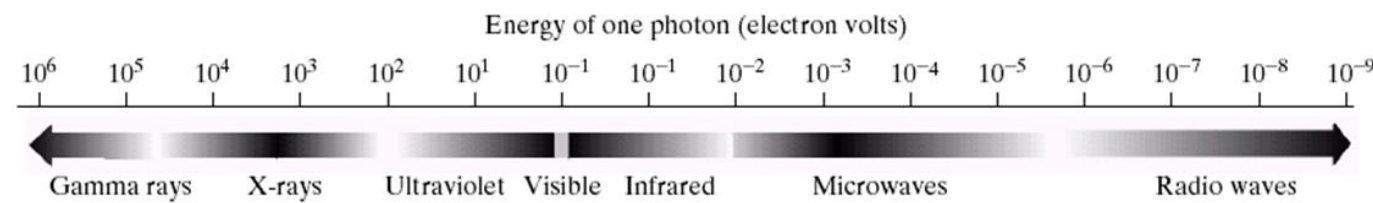


FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

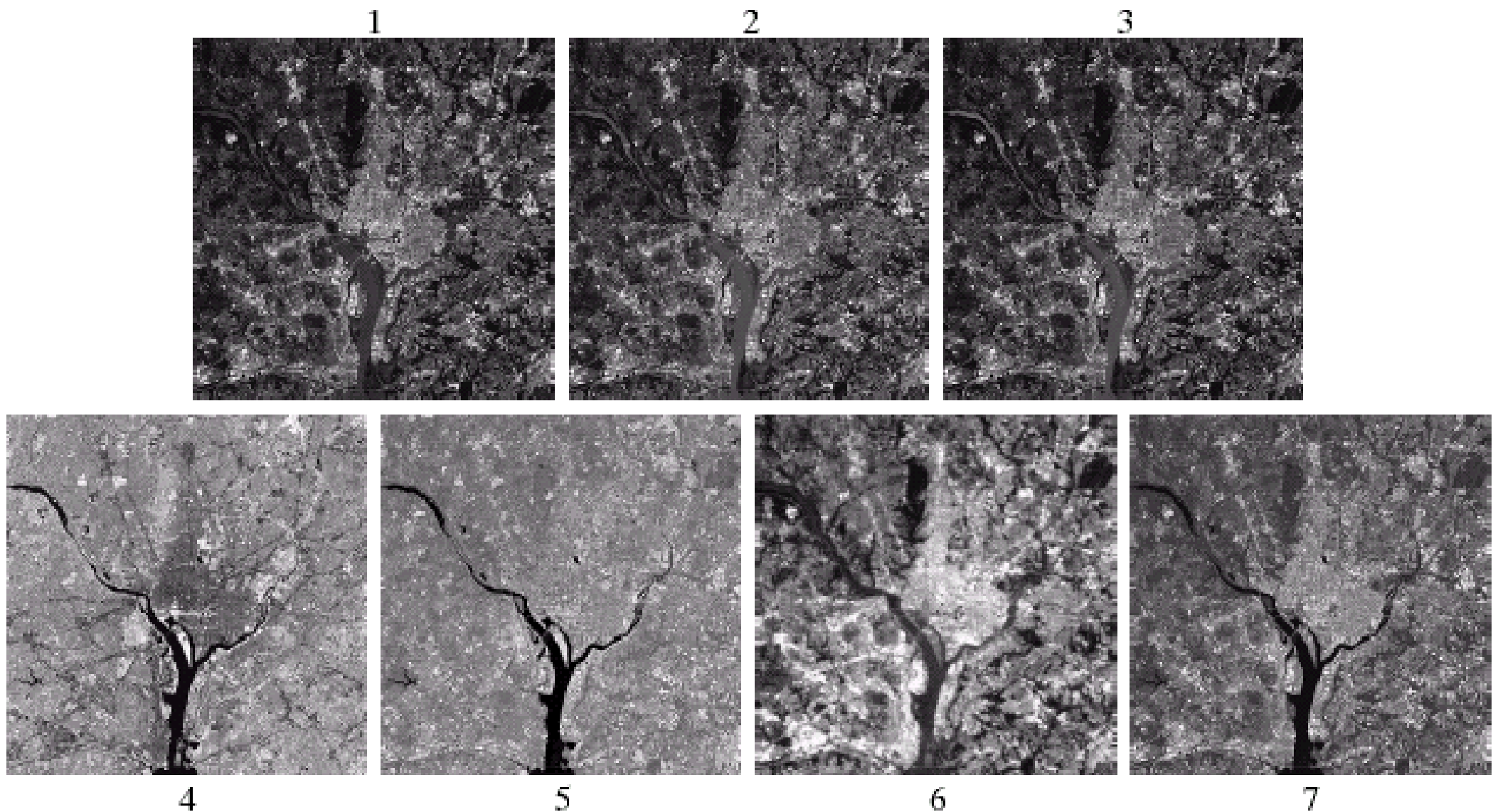
**TABLE 1.1**

Thematic bands of NASA's LANDSAT satellite.



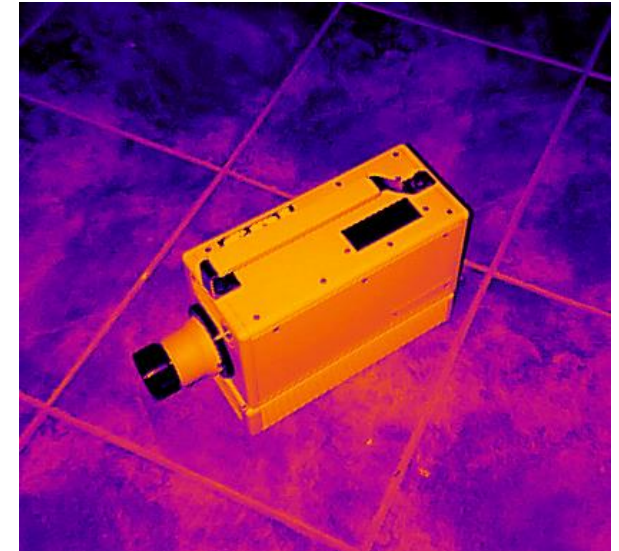
NASA LANDSAT satellite

| Band No. | Name                | Wavelength (μm) | Characteristics and Uses          |
|----------|---------------------|-----------------|-----------------------------------|
| 1        | Visible blue        | 0.45–0.52       | Maximum water penetration         |
| 2        | Visible green       | 0.53–0.61       | Measures plant vigor              |
| 3        | Visible red         | 0.63–0.69       | Vegetation discrimination         |
| 4        | Near infrared       | 0.78–0.90       | Biomass and shoreline mapping     |
| 5        | Middle infrared     | 1.55–1.75       | Moisture content: soil/vegetation |
| 6        | Thermal infrared    | 10.4–12.5       | Soil moisture; thermal mapping    |
| 7        | Short-wave infrared | 2.09–2.35       | Mineral mapping                   |



**FIGURE 1.10** LANDSAT satellite images of the Washington, D.C. area. The numbers refer to the thematic bands in Table 1.1. (Images courtesy of NASA.)

# Infrared for Night Vision

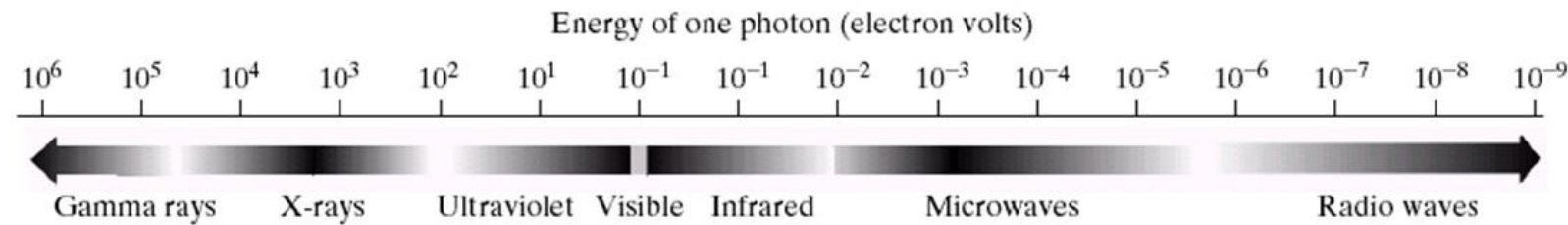


# Infrared for Thermal Imaging

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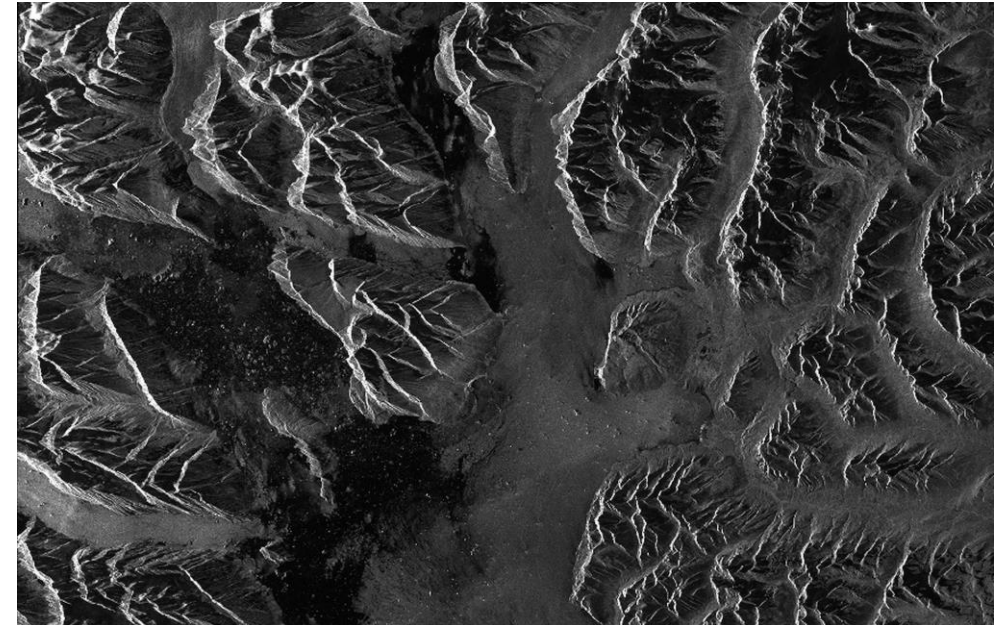
# Microwave Imaging



**FIGURE 1.5** The electromagnetic spectrum arranged according to energy per photon.

The dominant application of imaging in the microwave band is **radar** that can explore **inaccessible regions**.

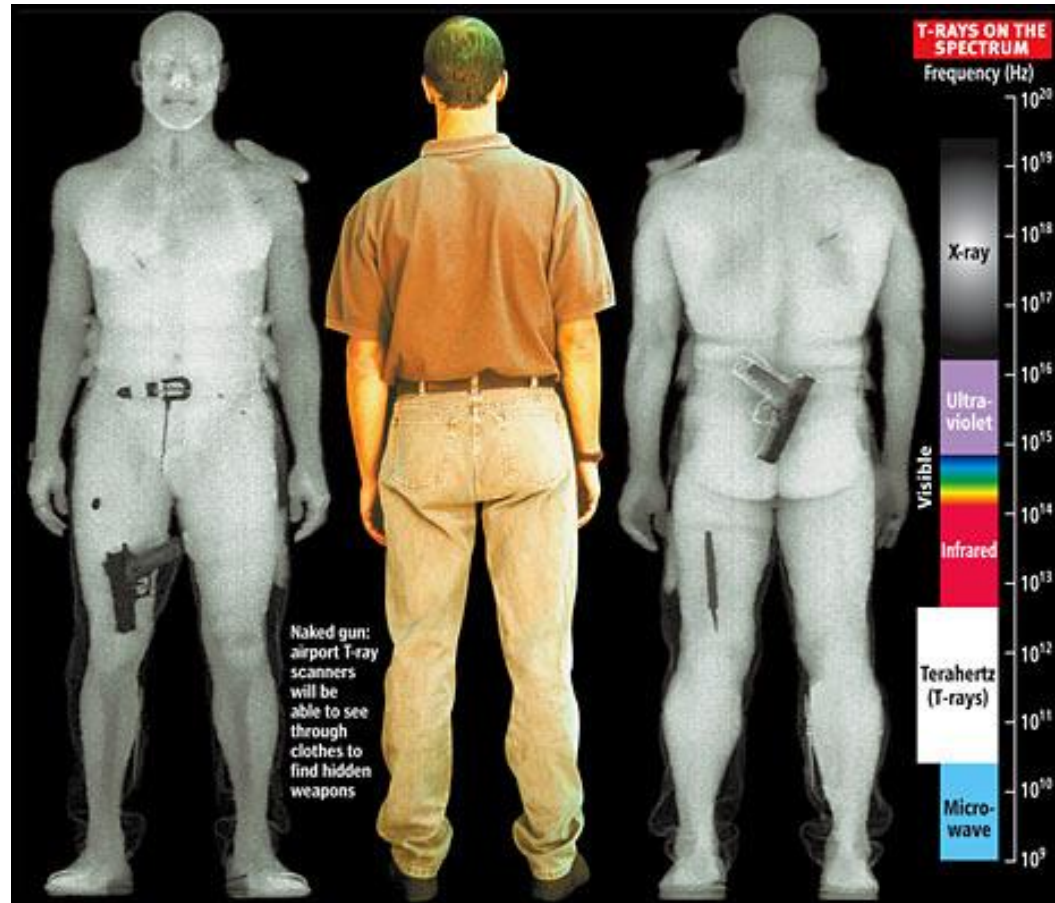
- The unique feature of imaging radar is its ability to collect data over virtually any region at any time, **regardless of weather or ambient lighting conditions**.
- An imaging radar works like a flash camera in that it provides its own illumination (microwave pulses) to illuminate an area on the ground or an object and take a snapshot.
- A radar uses **an antenna and computer** to record its image by collecting reflected microwave energy.



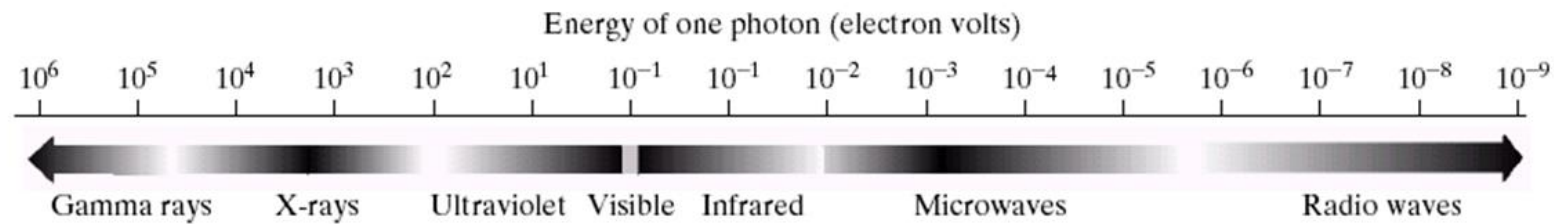
ERS-2 Greenland

<https://www.cgg.com/geoscience/satellite-mapping/radar-satellite-imagery>

# Airport Full-Body Scan



# Radio Band Imaging

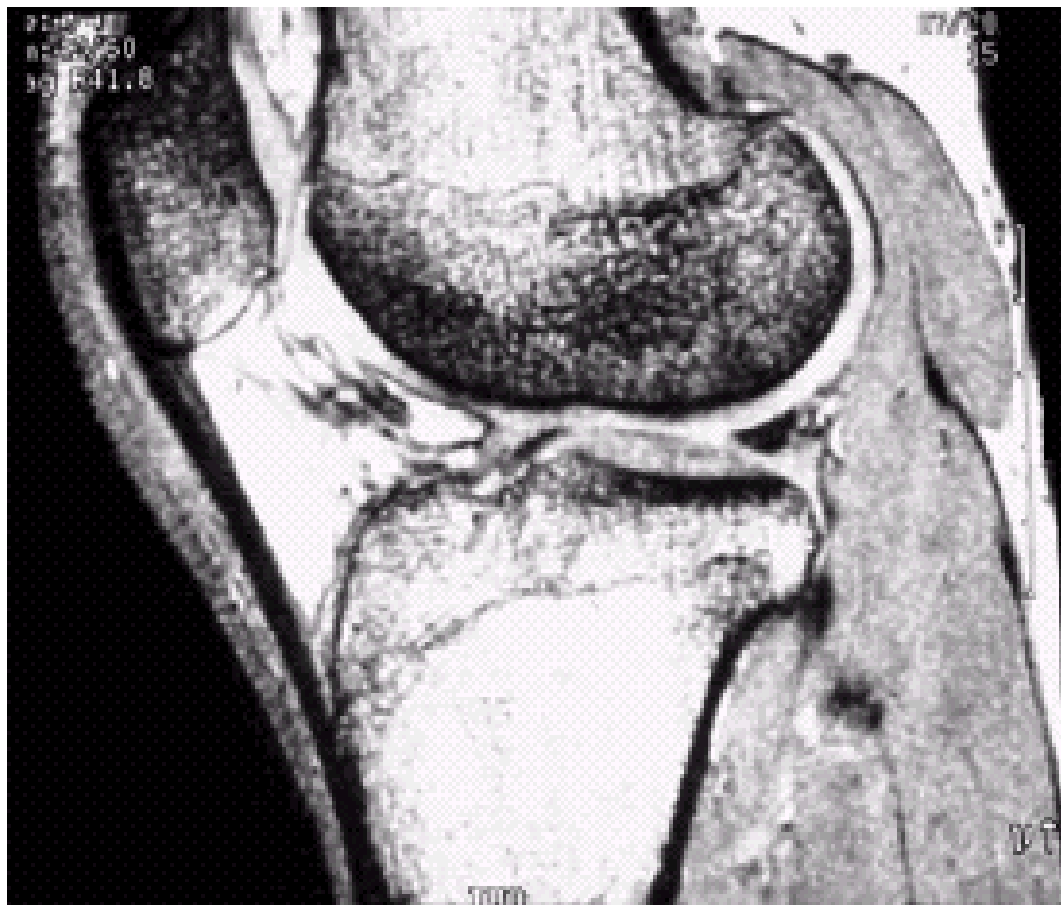


**FIGURE 1.5** The electromagnetic spectrum arranged according to energy per photon.

In medicine, radio waves are used in **Magnetic resonance imaging (MRI)** .

- MRI technique places a patient in a powerful magnet and passes radio waves through his or her body in short pulses.
- Each pulse causes a responding pulse of radio waves to be emitted by the patient's tissues.
- Different tissue types with different water densities and thermodynamics are reflected by different energy emitted by the tissue.
- The location from which these signals originate and their strength are determined by a computer.



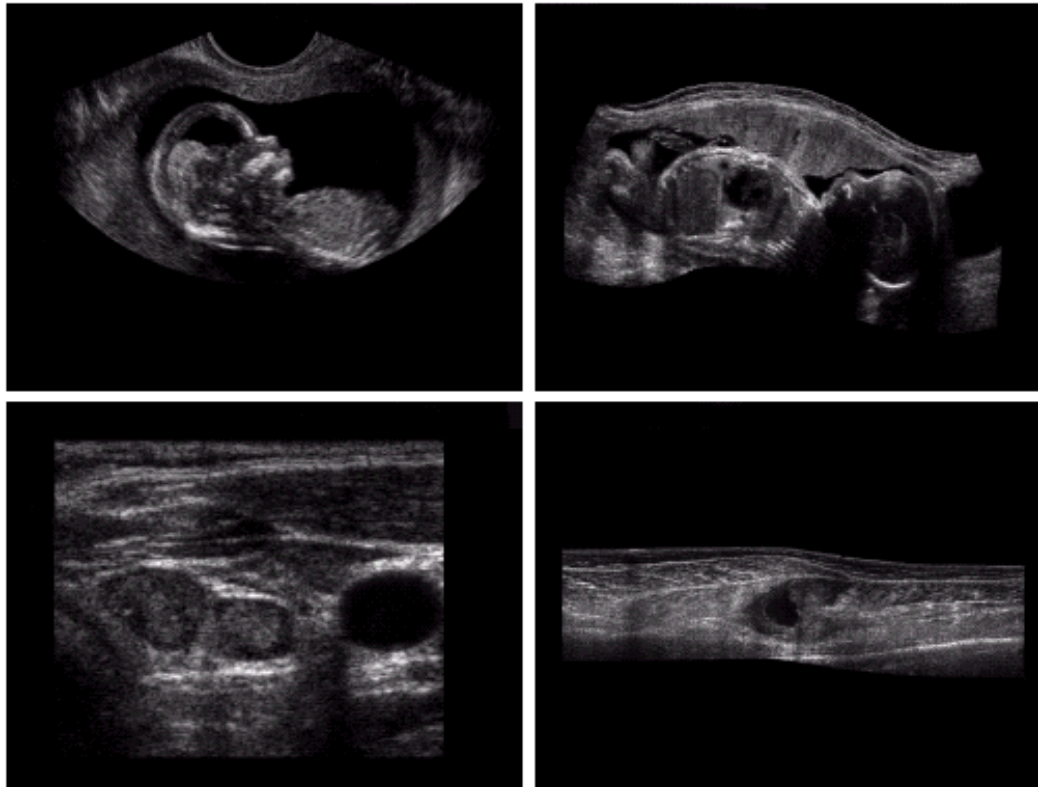


a b

**FIGURE 1.17** MRI images of a human (a) knee, and (b) spine. (Image (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

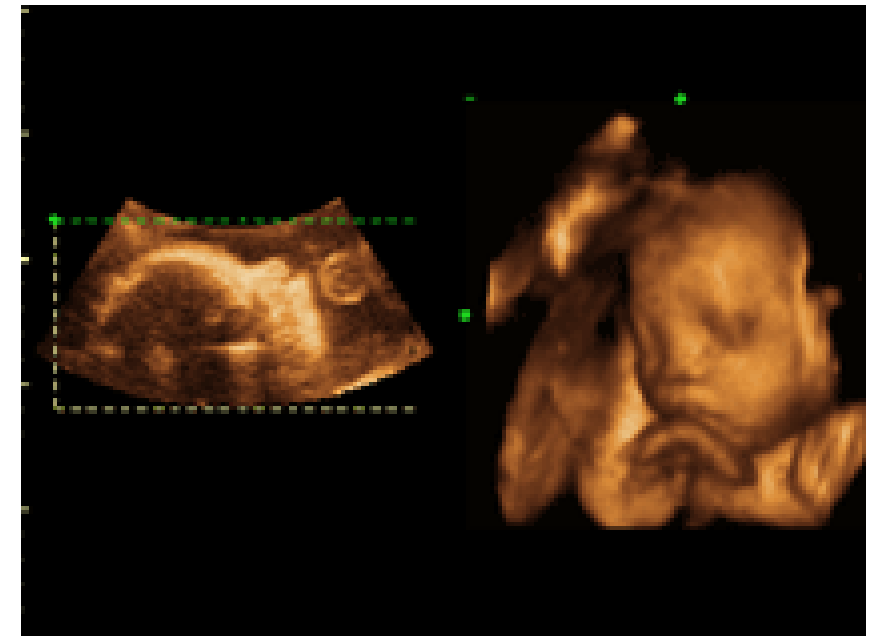
# Other Modalities: Ultrasound

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a b  
c d

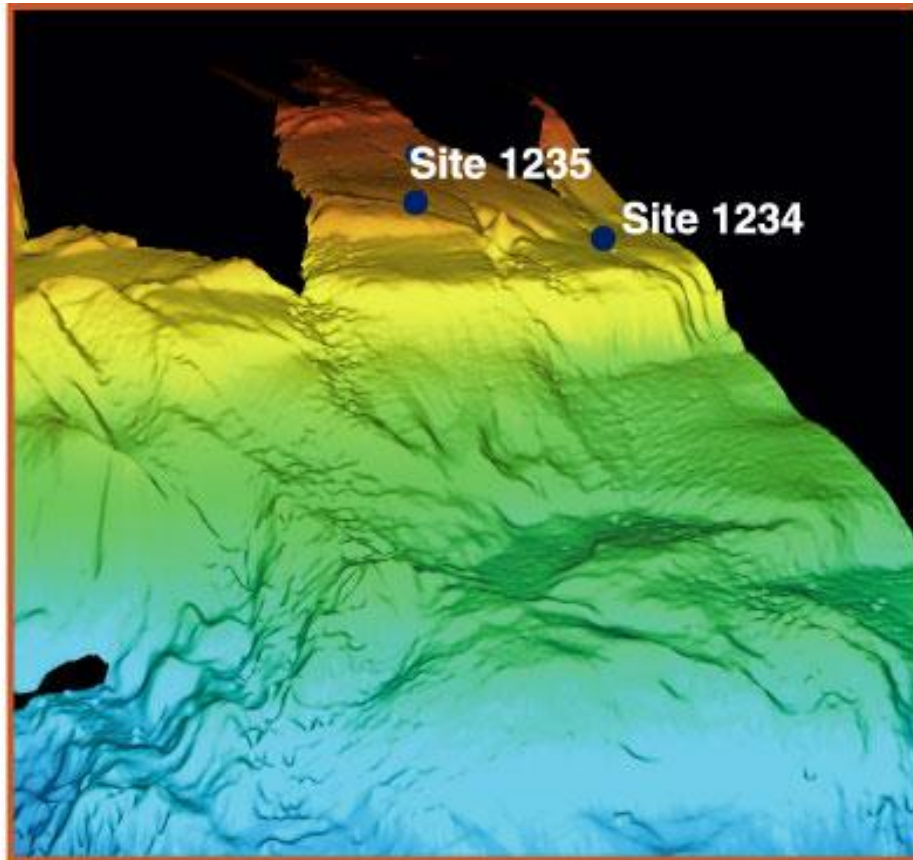
**FIGURE 1.20**  
Examples of  
ultrasound  
imaging. (a) Baby.  
(2) Another view  
of baby.  
(c) Thyroids.  
(d) Muscle layers  
showing lesion.  
(Courtesy of  
Siemens Medical  
Systems, Inc.,  
Ultrasound  
Group.)



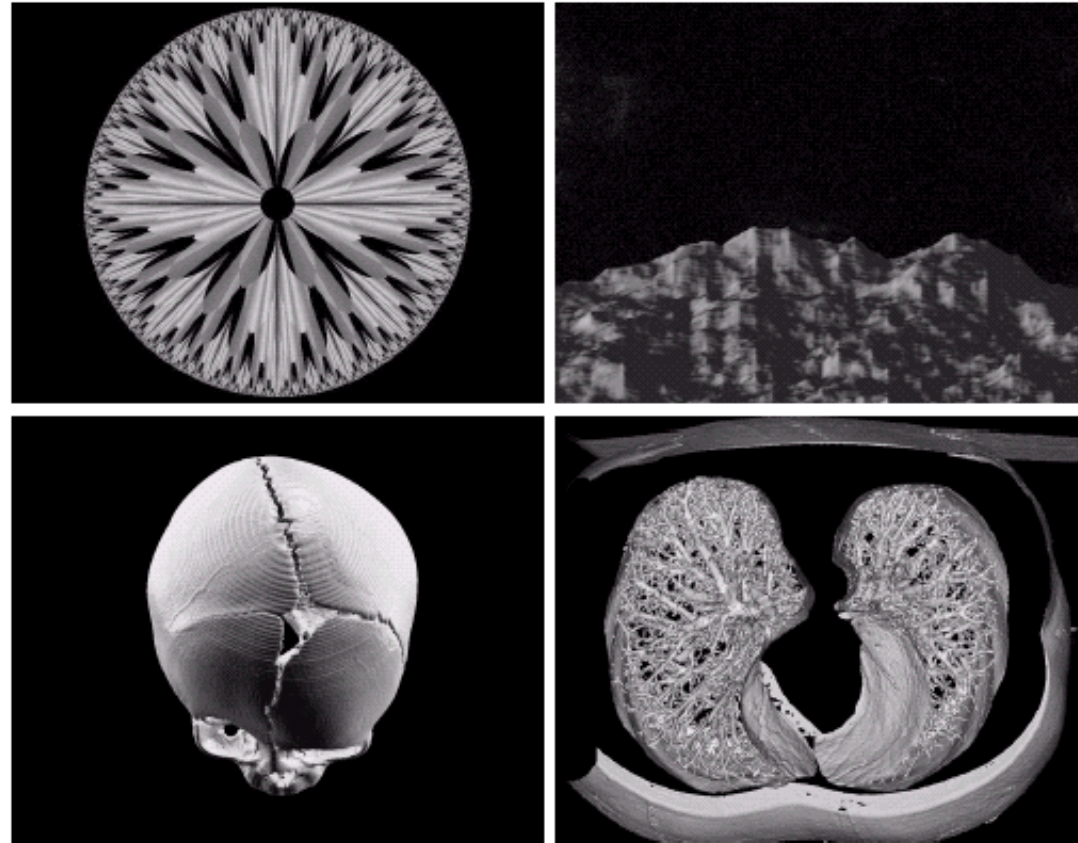
# Other Modalities:

## Acoustic Imaging and Computer Graphics (CG)

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Seafloor Sonar Mapping



**FIGURE 1.22**  
(a) and (b) Fractal images. (c) and (d) Images generated from 3-D computer models of the objects shown. (Figures (a) and (b) courtesy of Ms. Melissa D. Binde, Swarthmore College, (c) and (d) courtesy of NASA.)

# Other Modalities: Generative AI

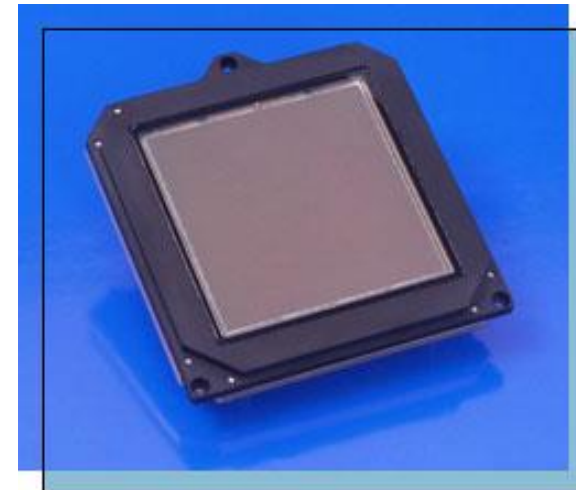
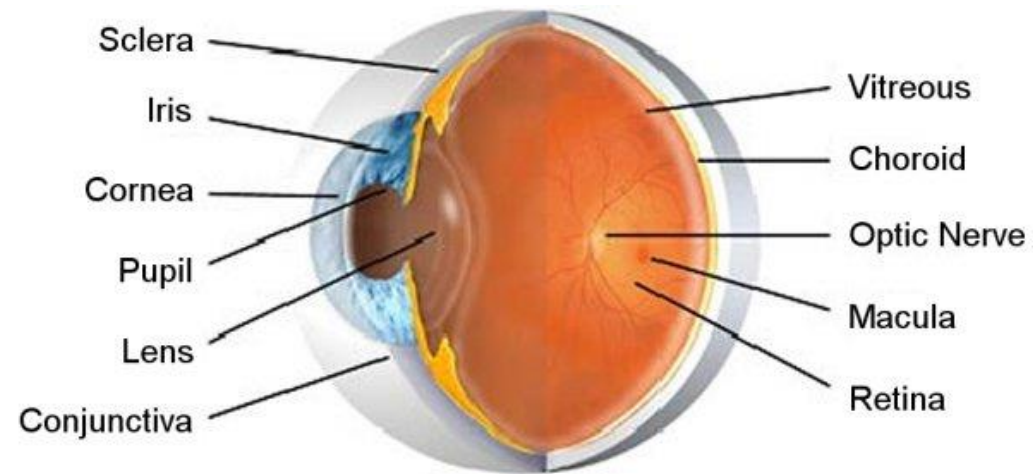


# Questions

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How does the human eyes perceive from real-world?

What kind of properties of human visual perception can help us optimize the imaging system ?



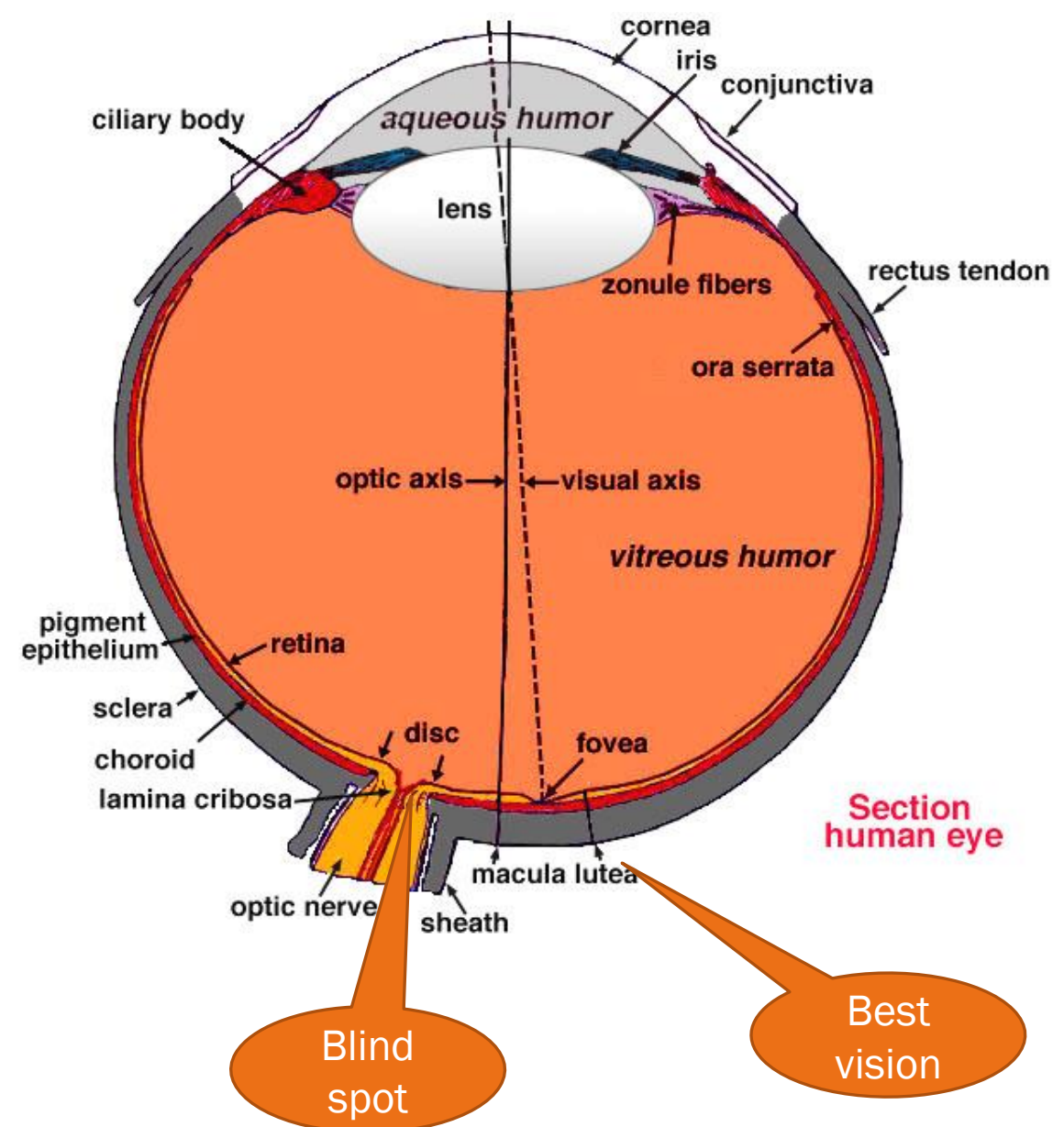
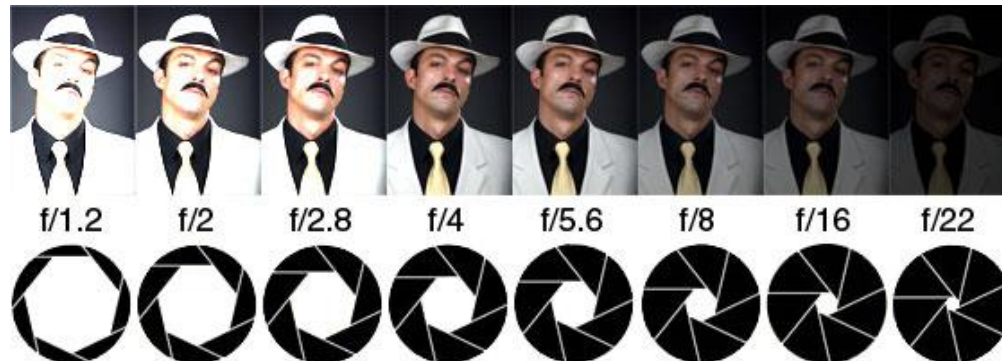
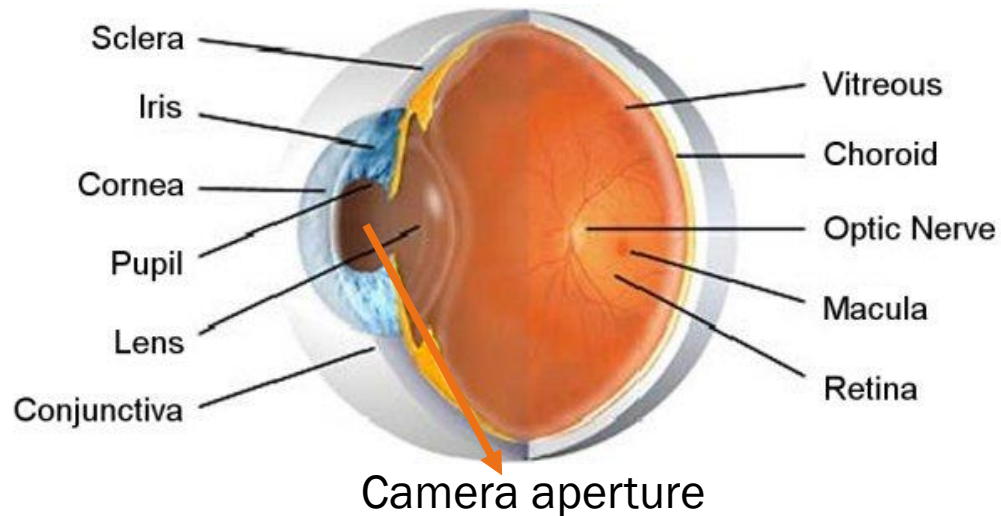
# What is this image?

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The human vision system (HVS) is a very complex and intelligent system composed of both the eye and brain.

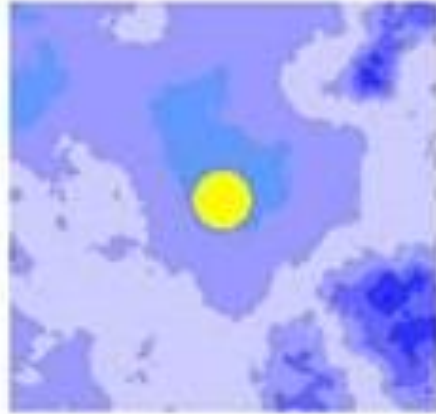
# Eye Structure



<http://www.pambaumann.com/blog/2014/8/26/aperture-and-its-role-in-the-exposure-triangle>

# Blind Spot

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1 2 3 4 5 6

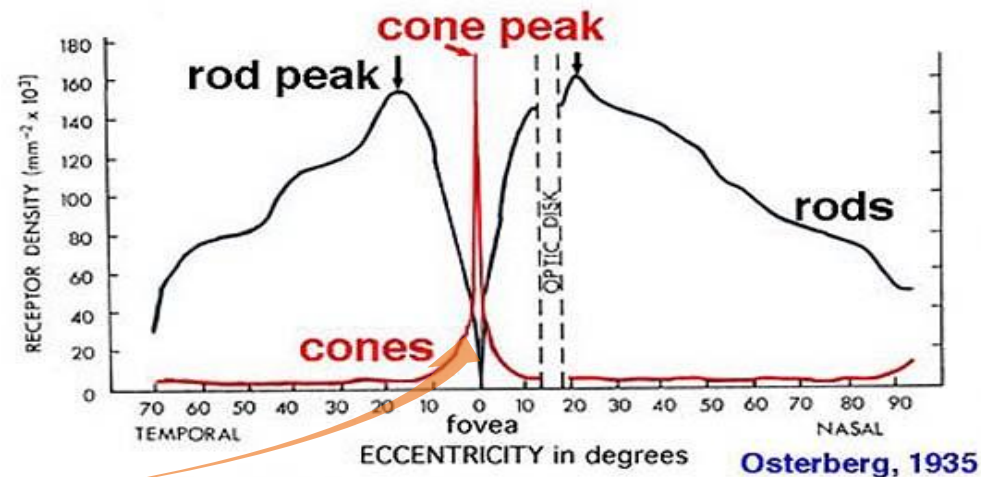
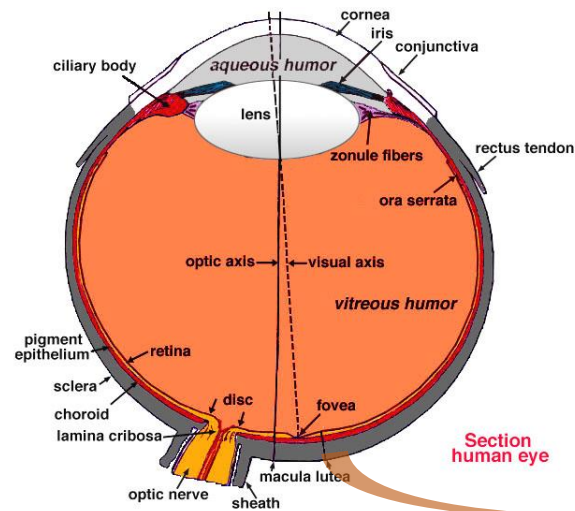
## Blind Spot (Optic Disk)

Close your right eye and look directly at the number 3. Can you see the yellow spot in your peripheral vision? Now slowly move towards or away from the screen. At some point, the yellow spot will disappear.

# Human Eyes: Retina

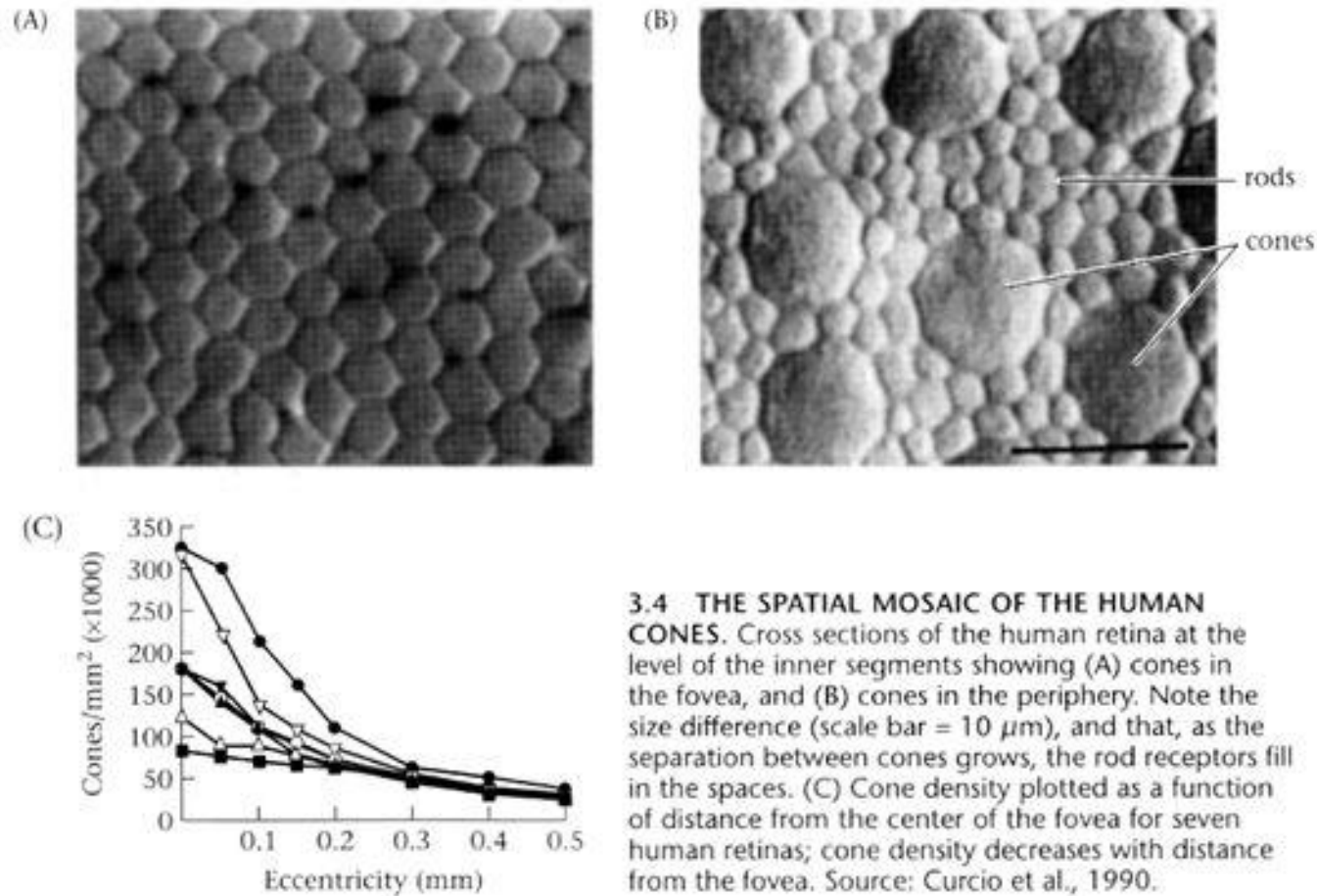
The **retina** is analogous to the digital camera containing discrete sensing elements known as **rods** and **cones**.

- **Rods** (100M): long and thin, low light vision.
- **Cones** (6.5M): short and thick, photopic vision, color vision, around fovea.



# Human Eyes: Rods and Cones

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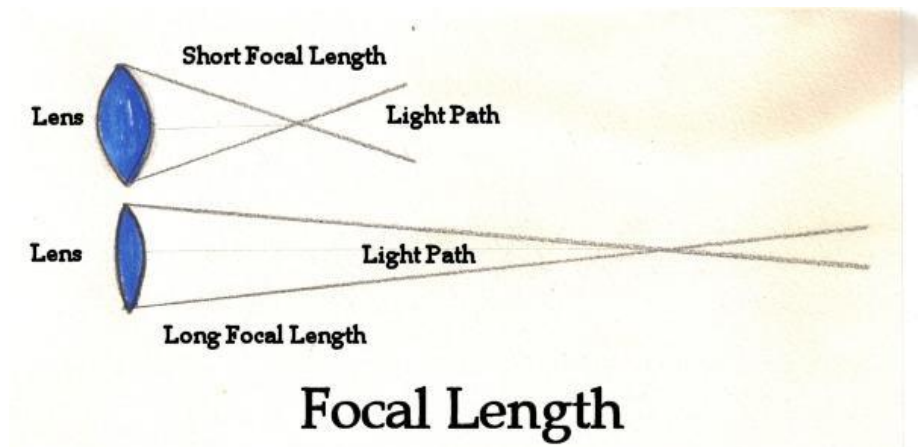
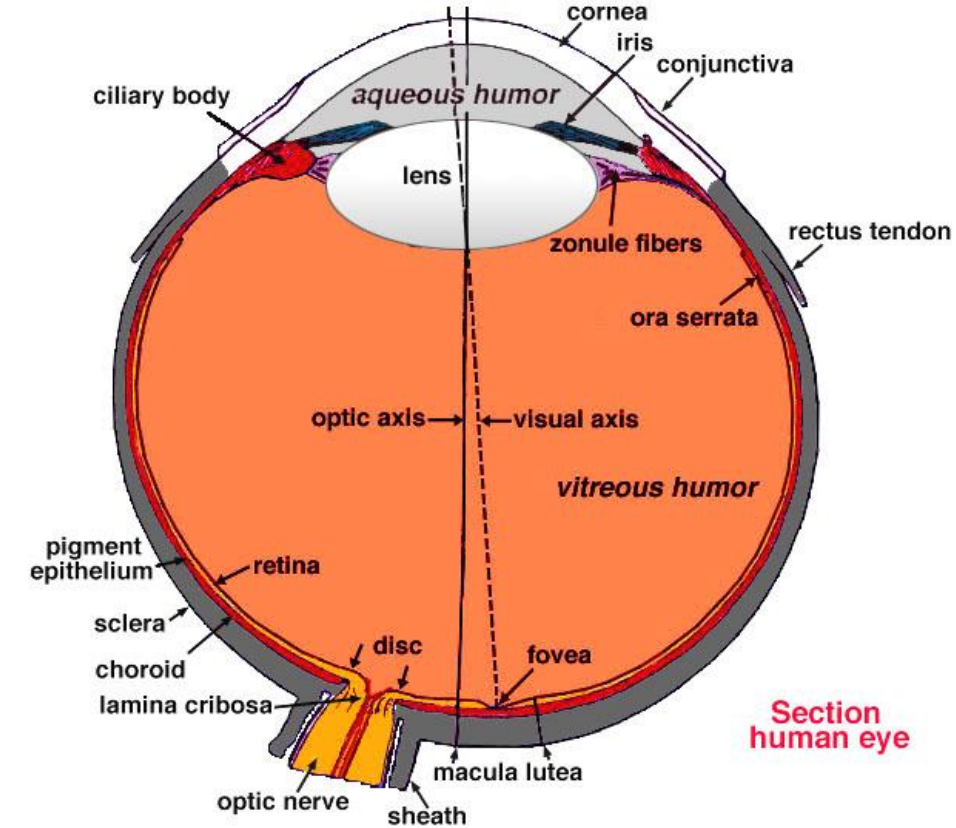


# Human Eyes: Lens

The principal difference between the **lens of the eye** and **ordinary optical lens** is that the former is flexible.

The shape of the lens is controlled by tension in the fibers of the **ciliary body**.

- When lens are controlled to be flattened, we can focus on distant objects.
- When lens are controlled to become thicker, we can focus on object near the eye.



# Elements of Visual Perception

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## Light

- Received from an object in the real-world

## Luminance and Intensity

- Collected by the human eye, luminance is weighted by eye's sensitivity to wavelength

## Brightness and Contrast (subjective)

- Perceived by the human vision system

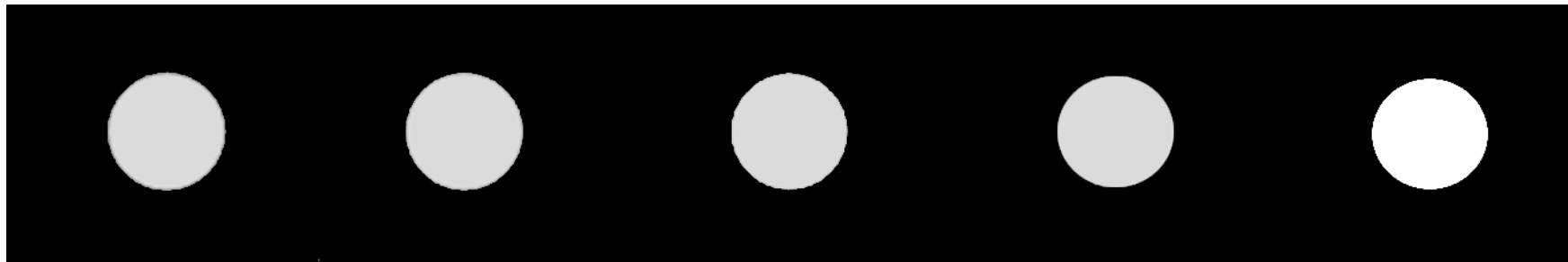
Practical implication: It helps us know how many gray levels are needed for image representation under different environments

# Brightness and Contrast

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**Brightness (subjective brightness):** Perceived luminance depending on the luminance of surround.

- One object with different surroundings could have different brightness.
- Two objects with different surroundings could have identical *brightness* but different *luminance*.



- The HVS is sensitive to **luminance contrast** rather than the **absolute luminance values** themselves.

# Weber's Law

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**Weber's law:** a just noticeable difference in light intensity is a constant proportion of the background intensity.

$$\frac{|f_s - f_0|}{f_s} = \text{constant} \quad \begin{array}{l} f_0 \text{ (luminance of an object)} \\ f_s \text{ (luminance of its surrounding)} \end{array}$$

$\downarrow$

$$\frac{\Delta f}{f} \cong \Delta c \approx 0.02$$

$\Delta f$  is small for just noticeably different luminance.

We can determine the different gray levels as follows:

$$\frac{f_{k+1} - f_k}{f_k} = \Delta c \quad \rightarrow \quad f_{k+1} = f_k(1 + \Delta c)$$

- The number of gray levels  $N$  is determined by the Contrast Ratio (CR)  $\frac{f_N}{f_0}$  (where  $f_N$  and  $f_0$  are the brightest/darkest gray levels) and the sensitivity ratio  $\Delta C$ .

# Number of Gray Levels

Contrast ratio (CR) refers to the ratio of intensities between the most intense (brightest,  $f_N = 1$ ) and least intense (darkest,  $f_0 = \frac{1}{CR}$ ) elements of a scene, an image, or a display.

|                                 | Contrast sensitivity |                  |                   |
|---------------------------------|----------------------|------------------|-------------------|
|                                 | 2%                   | 1%               | 0.5%              |
| Contrast ratio                  |                      |                  |                   |
| 10<br>(office environment)      | 116<br>(~7 bits)     | 231<br>(~8 bits) | 462<br>(~9 bits)  |
| 30<br>(living room, television) | 172                  | 342              | 682               |
| 100<br>(cinema theater)         | 232<br>(~8 bits)     | 463<br>(~9 bits) | 923<br>(~10 bits) |

$$\square \quad f_N = 1 = \frac{(1 + \Delta c)^N}{CR} \rightarrow CR = (1 + \Delta c)^N$$

$\vdots$



$$f_{k+1} = f_k (1 + \Delta c)$$

$\vdots$



$$f_2 = \frac{(1 + \Delta c)^2}{CR}$$



$$f_1 = \frac{1 + \Delta c}{CR}$$



$$f_0 = \frac{1}{CR}$$

Number of gray levels:

$$N = \frac{\log(CR)}{\log(1 + \Delta c)}$$



|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 2.21**

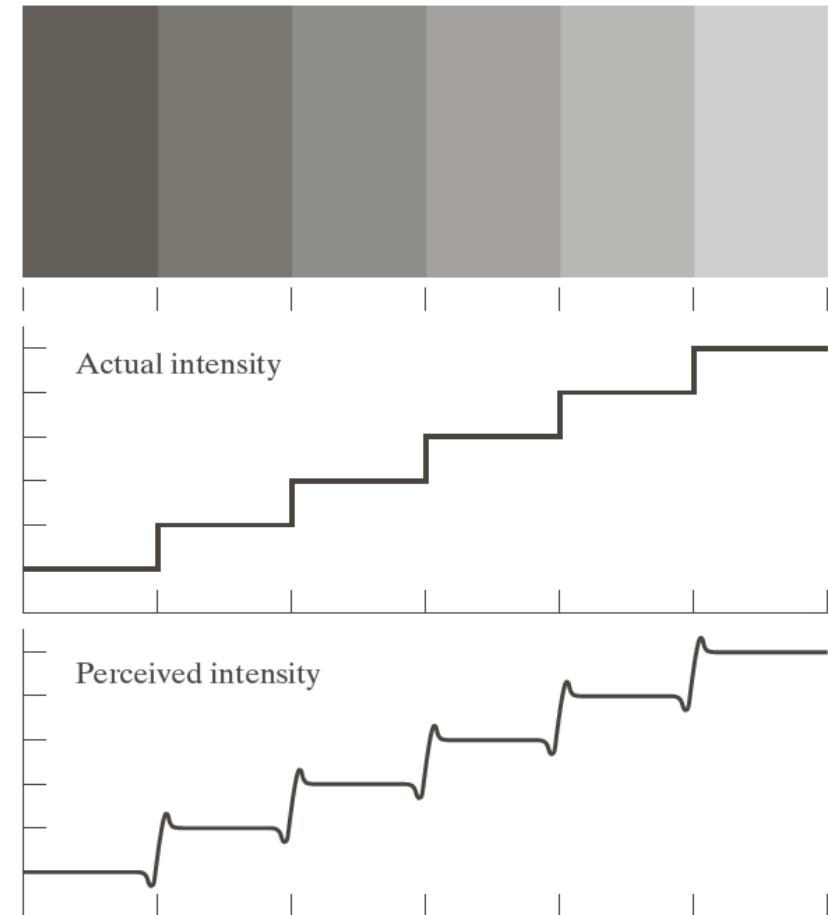
(a)  $452 \times 374$ , 256-level image. (b)–(d) Image displayed in 128, 64, and 32 gray levels, while keeping the spatial resolution constant.

# Mach Bands

The HVS tends to undershoot or overshoot around the boundary of regions of different intensities.

a  
b  
c

**FIGURE 2.7**  
Illustration of the Mach band effect. Perceived intensity is not a simple function of actual intensity.



# Visual Illusion

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|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 2.9** Some well-known optical illusions.

